

FAULT CODE 234 - Engine Crankshaft Speed/Position - Data Valid But Above Normal Operating Range - Most Severe Level

Warnings and Cautions

⚠ WARNING ⚠

DO NOT OPERATE A DIESEL ENGINE WHERE THERE ARE OR CAN BE COMBUSTIBLE VAPORS. These vapors can be sucked through the air intake system and cause engine acceleration and overspeeding that can result in a fire, an explosion, and extensive property damage. Numerous safety devices are available, such as air intake shutoff devices, to minimize the risk of overspeeding where an engine, due to its application, might operate in a combustible environment, such as due to a fuel spill or gas leak. Remember, Cummins Inc. has no way of knowing the use you have for your engine. THE EQUIPMENT OWNER AND OPERATOR ARE RESPONSIBLE FOR SAFE OPERATION IN A HOSTILE ENVIRONMENT. CONSULT YOUR Cummins® AUTHORIZED REPAIR LOCATION FOR FURTHER INFORMATION.

⚠ CAUTION ⚠

If the engine speed stays above 3500 rpm, the engine must be shut off immediately or engine damage can occur.

⚠ CAUTION ⚠

To reduce the possibility of damaging a new engine control module (ECM), all other active fault codes must be investigated prior to replacing the ECM.

Troubleshooting Summary

STEPS		SPECIFICATIONS
STEP 1.	Identify the reason for the overspeed.	

STEPS		SPECIFICATIONS
	STEP 1A. Check for proper operating conditions.	Motoring downhill?
	STEP 1B. Check for an alternate fuel source.	Alternate fuel source?
	STEP 1C. Check the engine rpm with INSITE™ electronic service tool.	Correct revolutions per minute (rpm) reading?
	STEP 1D. Check for an active fault at low engine rpm.	Inactive fault code at low rpm?
	STEP 1E. Test the vehicle.	Fault Code 234 inactive?
STEP 2.	Check ECM calibration and clear fault codes.	
	STEP 2A. Check if an ECM calibration update is available.	If a calibration update for this fault code is available, does the ECM contain that revision or higher?
	STEP 2B. Disable the fault code.	Fault code inactive?

STEP 1. Identify the reason for the overspeed.

STEP 1A. Check for proper operating conditions.

Conditions :

- Turn keyswitch OFF.

Action	Specification/Repair	Next Step
Check for proper operating conditions. • Check if the engine was motoring downhill when the fault was logged.	Motoring downhill? YES	1B
	Motoring downhill? NO Repair : Check for engine damage. • Electronics are ok. • Check the base engine for damage due to overspeed condition.	2A

STEP 1B. Check for an alternate fuel source.

Conditions :

- Turn keyswitch OFF.

Action	Specification/Repair	Next Step
Check for an alternate fuel source. • Check if the driver reported a fuel-controlled event where the engine rapidly accelerated to the engine tachometer red area.	Alternate fuel source? YES Repair : Locate any alternate fuel sources, such as operating the engine near flammable vapors, blown turbocharger seals, etc.	2A
	Alternate fuel source? NO	1C

STEP 1C. Check the engine rpm with INSITE™ electronic service tool.

Conditions :

- Turn keyswitch ON.
- Engine is running.

Action	Specification/Repair	Next Step
Check the engine rpm with INSITE™ electronic service tool. Compare the engine speed reading on INSITE™ electronic service tool to a mechanical tachometer or the dash tachometer.	Correct revolutions per minute (rpm) reading? YES	1D
	Correct revolutions per minute (rpm) reading? NO Repair : Inspect the crankshaft engine speed sensor and camshaft engine position sensor as outlined in Fault Codes 689 and 778.	Troubleshoot Fault Codes 689 and 778.

STEP 1D. Check for an active fault at low engine rpm.

Conditions : Turn keyswitch ON.		
Action	Specification/Repair	Next Step
Check for active Fault Code 234 when the engine is not running above high idle speed. -	Inactive fault code at low rpm? YES	1E
	Inactive fault code at low rpm? NO Repair : Replace the ECM. Refer to Procedure 019-031 in the Associated Procedures Table.	2A

STEP 1E. Test the vehicle.

Conditions : Turn keyswitch ON.		
Action	Specification/Repair	Next Step
Test the vehicle in a road test or the vessel in a sea trial to determine if the overspeed condition is still present. -	Fault Code 234 inactive? YES	2A
	Fault Code 234 inactive? NO Repair : Return to troubleshooting steps or contact a Cummins® Authorized Repair Location if all steps have been completed and checked again.	1A

STEP 2. Check ECM calibration and clear fault codes.

STEP 2A. Check if an ECM calibration update is available.

Conditions :

- Connect all components.
- Connect INSITE™ electronic service tool.

Action	Specification/Repair	Next Step
Compare the ECM code and revision number in the ECM to the calibration revisions listed in the ECM Calibration Revision History for applicable changes related to this fault code. • Use INSITE™ electronic service tool to find the present ECM code and revision number in the ECM. The ECM code and revision number are found in the Calibration Information section of System ID and Dataplate in Features and Parameters.	If a calibration update for this fault code is available, does the ECM contain that revision or higher? YES	2B
	If a calibration update for this fault code is available, does the ECM contain that revision or higher? NO Repair : If necessary, calibrate the ECM. Refer to Procedure 019-032 in Section 19. (/qs3/pubsys2/xml/en/procedures/105/105-019-032.html)	2B

STEP 2B. Disable the fault code.

Conditions :

- Connect all components.
- Connect INSITE™ electronic service tool.

Action	Specification/Repair	Next Step
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Disable and clear the fault code. Operate the engine within the "Conditions for Clearing the Fault Code" found in the Overview section of this troubleshooting procedure.	Fault code inactive? YES	Repair complete.
	Fault code inactive? NO Repair : Return to the troubleshooting steps or contact a Cummins® Authorized Repair Location if all steps have been completed and checked again.	1A

Associated Procedures

Associated Procedures			
Procedure Title	Procedure Number	Service Model Name	Bulletin Number
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/336/336-019-031.html)	ISL8.9 CM2880 L112	4358493
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/164/164-019-031.html)	ISL9.5 CM2150 SN	4310608
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/209/209-019-031-tr.html)	ISB3.9 CM2220 B107	4310792
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/206/206-019-031-tr.html)	QSNT14 CM876 N102	4325993

Associated Procedures			
Procedure Title	Procedure Number	Service Model Name	Bulletin Number
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/218/218-019-031.html)	QSB4.5 CM2150 B108	4326163
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/216/216-019-031.html)	QSB6.7 CM2150 B109	4326168
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/331/331-019-031.html)	QSB7 CM2880 B117	4358390
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/177/177-019-031-tr.html)	ISG11 CM2880 G106	4332695
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/177/177-019-031-tr.html)	ISG12 CM2880 G107	4332690
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/325/325-019-031-tr.html)	ISG11 CM2880 G108	4332901
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/325/325-019-031-tr.html)	ISG12 CM2880 G109	4332906
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/284/284-019-031-tr.html)	QSF2.8 CM2880 F104	4332741

Associated Procedures			
Procedure Title	Procedure Number	Service Model Name	Bulletin Number
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/284/284-019-031-tr.html)	QSF2.8 CM2880 F108	4332746
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/389/389-019-031-tr.html)	QSF3.8 CM2880 F112	4383825
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/379/379-019-031.html)	ISB/ISD6.7 CM2880 B126	4383693
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/383/383-019-031-tr.html)	ISB5.9 CM2880 B127	4383645
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/403/403-019-031.html)	QSL9.3 CM2880 L113	4383811
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/387/387-019-031.html)	ISF3.8 CM2220 F116	4383664
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/436/436-019-031.html)	QSC8.3 CM2880 C102	4388785
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/451/451-019-031.html)	QSB5.9 CM2880 B139	4388870

Associated Procedures			
Procedure Title	Procedure Number	Service Model Name	Bulletin Number
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/449/449-019-031.html)	QSB3.9 CM2880 B138	5411050
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/506/506-019-031.html)	ISD6.7 CM2880 D101	5411372
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/497/497-019-031.html)	ISF2.8 CM2220 F129	5411325
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/501/501-019-031.html)	ISF4.5 CM2220 F123	5411320
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/417/417-019-031.html)	QSG12 CM2880 G112	4388731
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/556/556-019-031.html)	ISF3.8 CM2220 F134B	5504165
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/577/577-019-031.html)	Z14 CM2670 Z103B	5504577
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/546/546-019-031.html)	D6.7 CM2670 D102B	5504515

Associated Procedures			
Procedure Title	Procedure Number	Service Model Name	Bulletin Number
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/629/629-019-031.html)	B6.2 CM2670 B156B	5579510
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/624/624-019-031.html)	X12 CM2670 X121B	5504455
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/622/622-019-031.html)	L9 CM2670 L128B	5504589

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